What is SAE International?

- SAE stands for Society of Automotive Engineers.
- It is a global association of more than 1,28,000 engineers and technical experts in the field of aerospace and automotive industries for the benefit of society.

8.2.1 Viscosity

Viscosity of a liquid is the ability that describes a fluid's resistance to flow. Oils with high density are highly viscous in nature. Viscosity decreases with increase in temperature of lubricating oil and increases with decrease in temperature. Society of Automotive Engineers has classified the lubricating oils based on the nature of viscosity. Viscosity of the oil increases with increase in SAE unit. Viscosity is more important among the properties of lubricating oil. It is measured by using the instrument called as viscometer. Viscosity is always measured along with the temperature. For example: SAE 40 at 210° F. This represents that the oil at 210° F has 40 units of viscosity. The higher the viscosity, the higher the SAE viscosiy grade number is.

8.2.2 Oiliness or Adhesiveness

Oiliness and adhesiveness are the properties of making an oily layer at the surfaces of metals in contact. This property should be high for lubricating oil. It is helpful in making the thin oil layer even at high temperatures in minute gaps.

8.2.3 Fluidity

Fluidity is the property of easy flow of lubricating oil in very small gaps. It helps in making a soft layer on the surfaces of junctions even at high temperature.

8.2.4 Volatility

Volatility is the ability of the oil to evaporate at high temperatures developed due continuous functioning of engine. Lubricating oil should have low volatility even at very high temperatures. Otherwise during the functioning of engine lubricating oil will evaporate which may lead to wastage of the lubricating oil, sometime even causing fire accidents.

8.2.5 Flash Point

The flash point is the minimum temperature, at which the oil produces the flash when an ignition source is brought close to the oil vapour and the flash will not continue when the ignition source is removed.

8.2.6 Fire Point

The fire point is the temperature at which the vapour of the oil continues to burn for atleast 5 seconds after ignition by open flame. The fire point will be generally at 10^0 to20^0C above the flash point. Lubricating oils should have higher fire point, to avoid evaporation and ignition.

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